

Introducing,

Intelligent
Battery
Systems

**Batteries that self-diagnose their problems
and report to you on their status
with no human intervention**

**Big Benefits on Uptime and
Savings on Manpower costs**

- Enables early preventive action.
- Ensures maximum power uptime.
- Eliminates unnecessary battery replacement.
- Monitor battery health and status remotely via Internet - get SMS/email alerts on specified threshold values.
- Huge savings on manpower costs as any number of battery banks can be monitored on a single screen.
- Makes warranty tracking easier by storing data like installation date, manufacturing date and battery capacity.

Ideally suited for applications like

- Mission critical data centers & server rooms
- Bank branches & ATMs • Telecom BTS and repeater stations • Power plants and sub stations
- Railway coaches & signalling systems
- Large steel plants • Manufacturing industries
- Solar / alternate energy systems installed with battery banks

- 50 service centers across India
- Part of a Rs.1300 crore group in the IT industry



RELICELL

Greenvision Technologies Pvt.Ltd

45/1, 4th Cross, Malleswaram, Bangalore - 560 003.
Tel: +91 080 4245 9458, Fax: +91 080 4245 9460.
Email: info@relicellbattery.com

Reader's Requests

I was one of EFY readers some 16-17 years ago during my college days, but lost touch. I have subscribed to EFY again about six months ago, and am happy to see that the magazine has retained all its old charms of beginner/hobbyist centric (mostly) and sometimes intermediate to complex circuit ideas, and loads of other informative articles. In short, I am enjoying the magazine quite a bit. My only requests are:

1. Include more of ATMEL AVR series and ARM processors based projects, built on low-cost development boards like Blueboard, Hawkboard, Beagleboard, mini2440 and Panda-board.
2. Publish projects based on tools and platforms like Arduino, Processing and Fritzting, detailing the production cycle—especially for moderately complex microelectronics projects.
3. Kits sold by EFY associates Kits'n'Spares should be made more affordable for young hobbyists.

Also, it is my wish to see EFY proactively reach out to colleges through roadshows, events, etc. A magazine like EFY wields tremendous power to make a significant difference to our education system, improving its orientation towards industrial applications. Hobby electronics is a great start, but we need students to go to the next level.

Banibrata

Through e-mail

EFY: It is wonderful to know that you have rejoined EFY readers' club. Welcome back! Your suggestions are very relevant and welcome. We are already working on some of them, and you should start noticing the changes soon.

Electronic Choke

Please publish the circuit and working of a 40W, 36W, 28W or 11W electronic choke.

Prabhat Kumar

Indore

EFY: You may refer to the '40W Electronic Ballast for Tubelights' circuit published in *Electronics Projects* Vol. 13. This book can be purchased from Kits'n'Spares.

Gas Leakage Alarm

Can you please provide the PCB layout for the 'Gas Leakage Alarm' circuit published in March issue? I have a 6-pin MQ-6 gas sensor. Can I use it in the given circuit?

Amit Ban

Bhopal

EFY: We do not provide PCB layouts for the circuits published under 'Circuit Ideas' section. You cannot use MQ-6 directly in the circuit. You need to amplify the MQ-6 output signal using a suitable amplifier circuit.

Motion Sensor for
Security Light

Regarding the 'Motion Sensor for Security Light' published in EFY Sep. 2010, I have following queries:

1. Why is resistor R3 used between the power supply and 12V, 2C/O relay? The relay doesn't energise due to R3.
2. What is the use of triac BT136 in the circuit?
3. Why is switch S1 connected to neutral line? I have connected this switch between the collector and the emitter of transistor T1, which is much safer for the circuit.

Bikash Kali Bhattacharjee

Kolkata

The author George Varkey replies:

1. Read RLI as a 5V, 2C/O relay. Resistor R3 is used as a current-limiting resistor.
 2. Triac BT136 was introduced to take the load current. In my original circuit, a home-made optocoupler (LED-LDR locally sourced) was used instead of the relay to fire the triac. The relay has been used here because an LDR with very high resistance in dark is not easily available. Also, the 'on' resistance of the LDR was large, which inhibited full output voltage in the load. The relay was found to be a good replacement with good isolation and physical size.
 3. The middle point of switch S1 should be connected to the emitter of transistor T1 and not to neutral line. Error is regretted.
- You can even remove capacitor C2 (1000µF) from the circuit to reduce part count and size of the PCB.